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Energy Self-Sufficient Greenhouse:
Photovoltaic-Powered Glasshouse and
Water System for Sustainable Plant Production

BACHELOR THESIS

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Klagenfurt, March 25, 2026

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Abstract

This bachelor's thesis deals with the sustainability and technical implementation of an energy-self-sufficient greenhouse designed for a year-round plant production in suburban environments. Considering global challenges, the integration of renewable energy and the trend toward urban self-sufficiency are increasingly moving into focus.

The aim of this work is to develop an autonomous system that uses organic photovoltaics (OPV) and a programmable watering system to make year-round cultivation without the need for external energy input possible. The methodological section provides a rationale for the selection of components and analyses them from both technical and ecological perspectives. Furthermore, the specific requirements of the plants are discussed in detail to ensure optimal growth. Additionally, an example of an ideal planting plan is presented to maximise the use of the limited space within the greenhouse.

An important aspect of the thesis is the simulation of the OPV energy yield, which was developed in Python. This simulation uses meteorological datasets for the location Klagenfurt and implements realistic weather conditions with suitable libraries. In this process, the energy yield was modelled for four different orientations (South, Southeast, East, and West) over an entire year.

The results show that a southeast orientation, with an annual yield of approximately 108.56 kWh, is the most efficient arrangement. Even with efficiency losses due to weather conditions, the system generates enough energy to power additional components. Finally, further approaches for expanding and deepening the research of this work are discussed.

Zusammenfassung

Diese Bachelorarbeit befasst sich mit der ökologischen Sinnhaftigkeit und technischen Umsetzung eines energieautarken Gewächshauses für die nachhaltige, ganzjährige Pflanzenproduktion im suburbanen Raum. Angesichts globaler Herausforderungen rückt die Integration von erneuerbarer Energien und der Trend zur urbanen Selbstversorgung zunehmend in den Fokus.

Das Ziel der Arbeit ist die Entwicklung eines autonomen Systems, das mittels organischer Photovoltaik (OPV) und eines programmierbaren Bewässerungssystems eine ganzjährige Bewirtschaftung ohne externe Energiezufuhr ermöglicht.

Im methodischen Teil wird die Auswahl der Komponenten begründet und hinsichtlich der Technik und der ökologischen Gesichtspunkte analysiert. Weiters wird auch genauer auf die Voraussetzung der Pflanzen eingegangen, um ein optimales Wachstum zu ermöglichen. Zudem wird auch ein Beispiel eines idealen Bepflanzungsplans dargestellt. Somit kann der kleine Platz im Gewächshaus optimal genutzt werden.

Ein zentraler Bestandteil ist die Simulation des Energieertrags der OPV, welche in Python entwickelt wurde. Diese Simulation nutzt meteorologische Datensätze des Standorts Klagenfurt und implementiert mittels geeigneter Bibliotheken realistische Wetterbedingungen. Dabei wurde der Energieertrag für vier verschiedene Himmelsrichtungen (Süd, Südost, Ost, West) über ein gesamtes Kalenderjahr modelliert.

Die Ergebnisse zeigen, dass eine Südostausrichtung mit einem Ertrag von ca. 108.56 kWh die effizienteste Konfiguration darstellt. Trotz witterungsbedingter Effizienzeinbußen reicht die erzeugte Energie aus, um eventuell weitere Komponenten zu betreiben. Am Ende werden weitere Ansätze geteilt, wodurch diese Arbeit vertieft werden kann.

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Chapter 1

Introduction

1.1 Motivation and Objectives

The idea for this project was inspired by my personal passion for plants and my interest in combining it with my technical know-how. Working with plants is not only a relaxing activity for me, but it also leads to a more conscious lifestyle. At the same time, I am fascinated by technical challenges, especially in the areas of energy efficiency and sustainable supply.

From this combination, the concept of a self-sufficient greenhouse was developed. The goal is to design a system that supports the year-round cultivation of fruits and vegetables without relying on external energy sources. A key aspect for me is the applicability to real-life situations: the concept should be designed in a way that it can be implemented in a typical suburban garden, making it accessible to a large share of the population.

The core of this work is therefore the technical development aiming for an energy-autonomous greenhouse. Furthermore, it will examine how much energy is required to operate such a system and whether a complete supply is feasible through photovoltaics. In addition, the study will analyse whether year-round self-sufficiency with fruits and vegetables is possible under these conditions.

1.2 Aim

This thesis aims to design and develop an energy-autonomous greenhouse that has a watering system. The system should be suitable for widespread use, particularly in suburban environments. In addition, the study considers different factors that influence the efficiency of solar energy, such as the orientation of the greenhouse.

The specific goals of this thesis are:

- First, I am going to design a concept for an energy-autonomous greenhouse that includes a watering system.

- Moreover, the whole project will be evaluated with the aspect of cost and sustainability.
- To analyse the energy requirements for year-round operation, we will consider the weather and the monthly average temperature.
- Also, I am going to simulate solar energy production under different real-world conditions, including various orientations of the greenhouse (e.g., south-facing, southeast-facing).
- Finally, some further adjustments will be discussed.

1.3 Structure

The structure of this thesis is designed to guide the reader through the development and evaluation of a self-sufficient greenhouse system. The first chapter introduces the motivation behind the project, outlines the objectives of the work, and discusses the relevance of combining sustainable energy systems with food production.

The second chapter focuses on the materials required for the construction of the greenhouse. Particular attention is paid to sustainability, while also considering cost factors to ensure that the system remains accessible and practical for everyday use.

In the third chapter, the photovoltaic system is analysed in detail. The basic functionality of the model used is explained, followed by four orientation scenarios.

- peak energy production
- south-facing orientation
- southeast-facing orientation
- east-facing orientation
- west-facing orientation

These scenarios are first investigated at a theoretical level. Consequently, the yield from different orientations will be simulated in Chapter Five.

The fourth chapter focuses on creating an ideal vegetable cultivation plan for a cold greenhouse. Furthermore, it covers the optimal environmental conditions essential for plants to grow, alongside the fundamental preparations required, such as soil quality and structure.

Chapter Five provides a detailed description of the simulation process. In particular, the simulation is about the production of the annual energy yield from different PV orientations under the given geographical conditions. The results of these simulations are compared and analysed to show the efficiency of the system under different real-world conditions, such as clouds.

Finally, the sixth chapter presents the conclusions drawn from the research. The paper discusses the profitability of the concept and outlines the potential for developing a real-life prototype in future work to test and validate the proposed system.

Chapter 2

Materials

2.1 Overview

In this project, many aspects must be included. One of them is the required materials. These are listed in this section, with each product followed by a brief summary of its key specifications. Furthermore, each item is evaluated in terms of sustainability, as this plays an important role in the overall concept. The entire greenhouse will be built and bought as a whole. The only elements not included in the calculation are the soil and the plant seedlings.

2.2 Components

2.2.1 Greenhouse

The Vitavia 'Playa 5900' was selected based on three key factors: a budget ceiling of approximately 1500€ for the greenhouse structure, the requirement to fit within a standard suburban garden plot, and the need for sufficient roof area to accommodate 8–10 OPV panels. With a footprint of 192×254 cm and a net cultivation area of 4.6 m^2 , the model satisfies all three constraints while remaining compact enough for smaller urban gardens. The single-slope design makes it easier to point the solar panels in the best production direction. Additionally, it is highly robust, because the structure is made of glass and the top (i.e. the roof) is constructed from hard plastic.[25] Furthermore, the greenhouse must be assembled manually, which requires a bit of skill and the support of volunteer helpers.

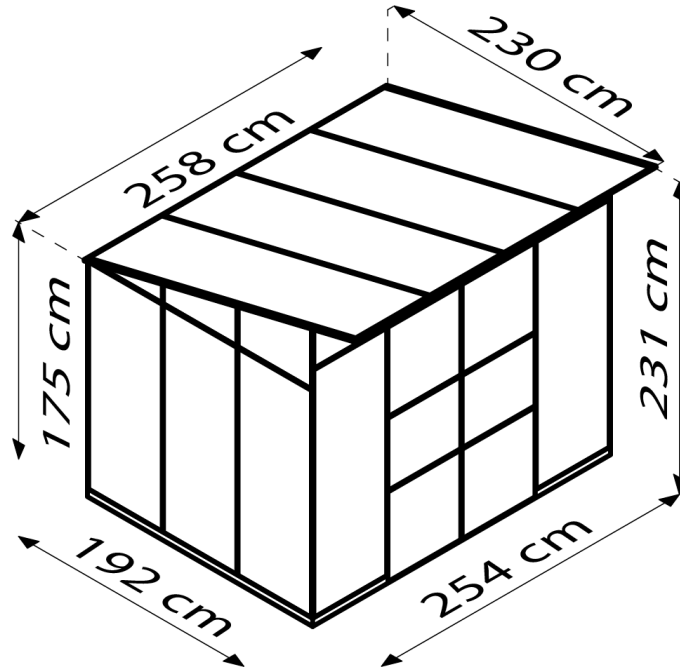


Figure 2.1: Greenhouse Playa 5900 (E.P.H. Schmidt and Co. GmbH[25])

2.2.2 Organic Photovoltaic (OPV)

The choice of this PV system was based on three primary factors: light transmission, structural stability, and environmental sustainability. The semi-transparent material of the modules ensures sufficient sunlight for the underlying flora. Furthermore, the low weight of the components optimises roof-load distribution. Compared to the standard silicon modules, organic photovoltaic is a more eco-friendly alternative. Because of this, the HeliaKit is used, but it is downsized because there will be only space for a maximum of 8 to 10 panels. Also, the Kit includes the needed cables and an inverter, which contains a Schuko socket. That means that the generated electricity can be directly fed into a local storage unit if it is not consumed. This makes the Kit a plug-and-play solution, which is easy to use. There is also an option to connect the Kit with the power grid, but this functionality is not intended for this project. The power management is handled by the APSystems EZ1 microinverter, a device specifically developed for small-scale urban solar installations.[10] The organic PV uses materials which are made of organic chemicals to produce electricity from sunlight. The production of energy is the same principle as for traditional PVs.[10] A more detailed theoretical explanation of how electricity is generated and how the process works is provided in Chapter 3.

2.2.3 Watering System

The decision to implement a battery-operated drip watering system was a result of the current unavailability of compact, programmable alternatives on the market. While a custom-built solution would have been technically possible, the primary goal was to ensure the concept remains simple and accessible to a wide audience. It is an application where a tubing, featuring perforations or small spray heads, is supplied with pressurised water. The drip line is laid out on the surface directly next to the plants. The water then drips onto the ground through these openings.[4]

The primary advantage of drip irrigation is the exact regulation of water flow. Accordingly, the water drips in smaller quantities over a longer period compared to traditional methods, such as using a watering can, where plants receive a large volume of water all at once.[7] To regulate this process, a watering supply controller is used, which can be programmed with various watering schedules and durations.

2.2.4 Sketch of the Greenhouse

I created a visual layout of the greenhouse to demonstrate the specific positioning of the PV system. Furthermore, I sketched the interior layout and estimated the available cultivation area based on these drawings. To determine the usable space, I applied the area formula $A = l \cdot b$. First, I calculate the total floor area, which is 48768 cm^2 . Then I computed the area of the entrance walkway (aligned with the door) as $A_2 = 61 \text{ cm} \cdot 50 \text{ cm} = 3050 \text{ cm}^2$. By subtracting A_2 from the total area, I determined the final net cultivation area of 4.5718 m^2 . For the calculation of the roof pitch, I utilised the tangent theorem as shown in the Sketch.

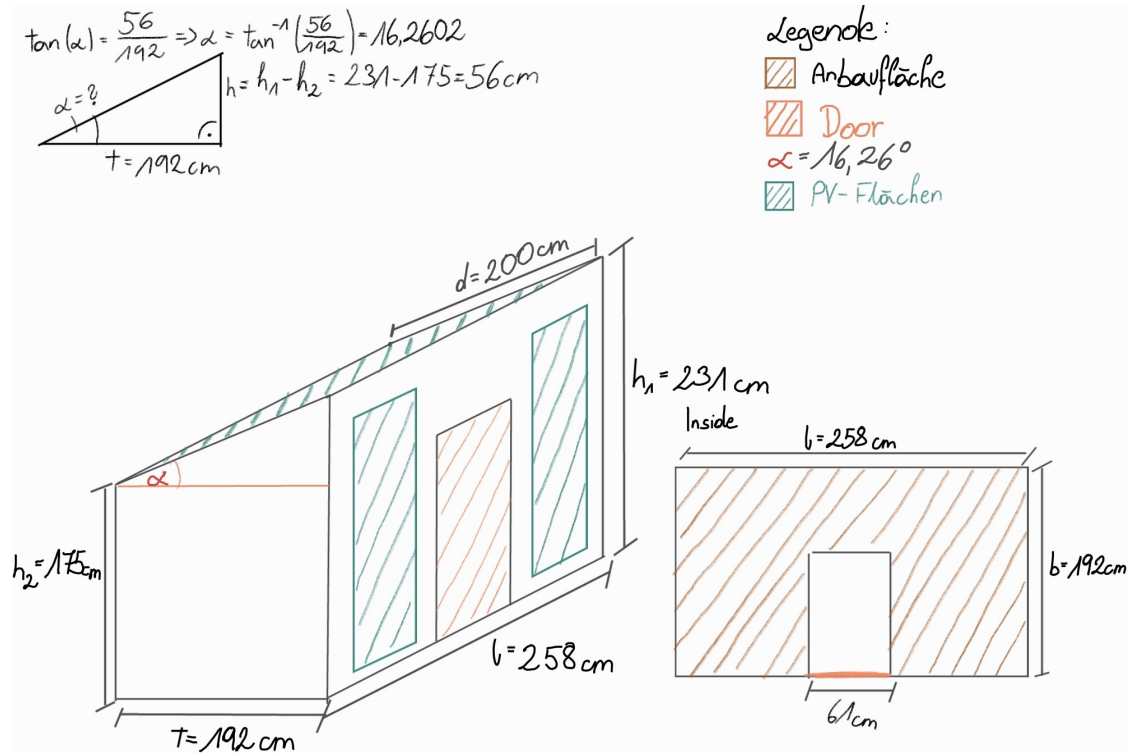


Figure 2.2: Sketch of the Greenhouse

2.3 Cost Breakdown and Financial Planning of the Project

The costs determined as part of this project refer to a real-life model that can be installed and practically used in a garden setting. For the calculation, the unit prices of the previously itemised components were used and scaled according to the required quantities. It should be noted that larger purchase quantities may lead to more profitable prices, which could have a positive impact on the overall project costs. In this case, the focus is on a small greenhouse with a surface area of 4.77 m^2 , where only relatively small quantities of materials are needed.

A central element of the planned energy supply is the so-called HeliaKit, which consists of 16 photovoltaic modules, an inverter, and the necessary connection cables. The estimated price for this complete system is 2390€, which was mentioned in a post from 2024. However, since this system is oversized for the intended project, it will be downsized. The price for the required components is based on the cost of the components in the large kit. However, this calculated kit is not available for sale because the company is still in its early development phase and currently focuses on selling to large companies.[10]

For this reason, I have also had an email correspondence with the company, and they informed me that this product will be available to private consumers in the coming years.

It should be mentioned that the prices indicated are variable and may change depending on market conditions and suppliers. The following cost calculation is based on the status as of February 2026.

Component	Quantity	Total Price (€)
Vitavia Greenhouse "Playa 5900"	1	1499
HeliaKit	1	1195
Gardena Drip Irrigation Set	1	49.99
Gardena Water Control Flex	1	40.99
Total		2785

Table 2.1: Estimated material costs for the self-sufficient greenhouse

2.4 Sustainability

There are many different ways to assess sustainability. In this work, the components are evaluated based on a specific set of criteria: energy efficiency, maintenance requirements, and service life, as well as manufacturer transparency. Regarding the greenhouse, the material composition is also taken into critical consideration.

2.4.1 Greenhouse

The Greenhouse consists of different materials, and its key components are being evaluated for sustainability.

Aluminium is considered a highly durable material due to its weather resistance and low maintenance requirements. It offers one of the highest recycling potentials among all recyclable materials. According to calculations by the Ellen MacArthur Foundation, every kilogram of recycled aluminum saves 8 kilograms of bauxite, 4 kilograms of chemicals, and approximately 14 kilowatt-hours of electricity.[19] However, despite this high potential, some material is unavoidably lost during each recycling cycle. Furthermore, the primary production of aluminium from bauxite is extremely energy-intensive and carries a significant carbon footprint, requiring 15 megawatt-hours of electricity to produce a single ton.[15]

The frame is also made of safety glass. The manufacturing of glass requires extremely high temperatures, making it a very energy-intensive process. On the positive side, the necessary raw materials, such as sand, lime, and soda, are natural and not risky. To improve the sustainability of the raw material supply, there is an increasing effort to use recycled glass. Additionally, glass is characterised by its exceptional durability and long service life. [9]

Unfortunately, the roof is made of polycarbonate (PC), a well-known type of plastic.

Plastics are frequently used because they are robust and weather-resistant. However, their recycling remains a highly sensitive and complex issue. There are several approaches to dealing with polycarbonate waste: Mechanical recycling often leads to "downcycling," where the material quality decreases. Alternatively, chemical recycling breaks the plastic down into its chemical building blocks, but this process, unfortunately, requires additional chemicals and high pressure. Consequently, from a sustainability perspective, this component is currently the least eco-friendly part of the structure.[2]

2.4.2 Organic Photovoltaics (OPV)

The sustainability of semi-transparent organic photovoltaics (OPV) is based on the application and durability. Historically, one of the primary criticisms of organic solar cells was their short lifespan. However, research has significantly improved this by implementing layout modifications and integrating a reductive interlayer. This interlayer ensures that the cells retain more than 84% of their original performance after 1000 hours of continuous irradiation.[30]

This represented the state of the research in 2023. The company Heliatek now offers a 20-year performance guarantee, although it provides a product guarantee of only 5 years. According to the company website, Heliatek prioritises sustainable manufacturing by excluding toxic heavy metals and rare earth elements from its production processes.[10] In 2023, tests showed that roofs of greenhouses equipped with semi-transparent OPV improved plant growth and got a higher survival rate compared to conventional glass roofs.[30]

2.4.3 Watering System

The watering supply controller contains different heavy metals due to the internal electronic circuitry. Furthermore, it is powered by a 9V alkaline battery, which primarily consists of zinc, iron, and manganese. These raw material resources are classified as harmful waste across Europe. [23]

The recycling process comprises the smelting of the batteries. However, a first material separation is attempted through crushing and sieving. The smelting process generates gases that require specialised filtration systems. After the smelting process, the materials are combined into an alloy.[23] Nevertheless, it can be concluded that the use of alkaline batteries is not eco-friendly due to the high complexity and energy requirements of the recycling process. Furthermore, the outer shell is made of hard plastic, which is processed using mechanical recycling methods. The material is first shredded and washed. It is then mixed with raw material in an exact ratio, depending on the desired properties of the final product. Unfortunately, separating the plastic back into its raw base materials is difficult, if not impossible. [24][17] While plastic is recyclable, it requires significant effort. However, in this specific case, it ensures a durable, long-lasting product.

Chapter 3

Photovoltaic

Why is photovoltaics becoming increasingly important? First of all, there are some obvious problems like climate change or Fossil Fuel Independence.[20] The solution to these problems is renewable energy, which can be wind power, biomass and photovoltaic. In this report, photovoltaics were selected because they can be used on a wide variety of surfaces, such as facades, rooftops, and open spaces, whereas the production of wind power and biomass requires significantly more land. Furthermore, the technical potential of PV is higher than that of the other two energy sources.[3]

3.1 How it Works: The Interplay of PV Components

The name "photovoltaics" is a combination of the Greek word for light and the unit 'volt'. Therefore, it could be translated as "light energy source". Photovoltaics can be briefly defined as the conversion of sunlight into electrical energy. The fundamental components of every photovoltaic system are the solar cells. These consist of silicon (a crystalline semiconductor). By adding foreign atoms (doping), a p-n junction is created, which in turn generates an electric field within the crystal. When sunlight hits the solar cell, charge carriers are released and directed to the outer contacts by the electric field. This process creates a voltage at the contacts. To achieve a higher voltage, multiple modules are connected in series.[20]

A simple setup of a photovoltaic system is shown in the image below. Several solar modules are connected in series and coupled to an inverter. This inverter converts the supplied direct current into alternating current and feeds it into the public grid. Additionally, there is a feed-in meter that measures the electricity supplied and a consumption meter that measures the electricity consumed within the household. However, nowadays, only surplus electricity is fed in, and mainly, self-consumption is the primary aim.[20]

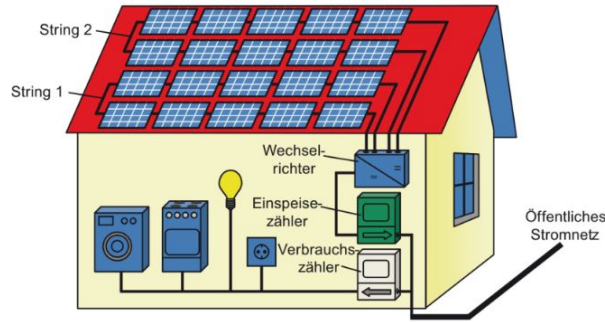


Figure 3.1: Setup of a classic Photovoltaic System (Picture 1.10 of the book [20])

3.2 Deep Dive: Organic Photovoltaics (OPV)

While conventional solar cells are based on inorganic silicon crystals, Organic Photovoltaics (OPV) use thin films of organic semiconductors (carbon-based molecules or polymers) to convert sunlight into electricity. A fundamental difference lies in the physics of light absorption. In silicon cells, light directly creates free electrons. In contrast, in organic materials, light absorption creates a very strongly bound electron-hole pair known as an exciton. Because of the high binding energy of these excitons, they do not automatically split into free charges. Instead, they must move through the material to a Donor-Acceptor heterojunction. This interface is the only location where the exciton can successfully be separated, transferring an electron to the acceptor and a hole to the donor to create an electrical current.[14]

The authors point out that the key factors for industrial interest in OPV are its unique physical properties. Because organic semiconductors can be processed at low temperatures from solutions or steam, they can be applied onto lightweight, flexible plastic substrates. This enables a wide range of new consumer products that are thin, bendable, and portable. Furthermore, the low-temperature manufacturing process significantly reduces the energy payback time compared to silicon-based cells, making OPV a highly sustainable energy source.[14]

Organic solar cells face a problem: the exciton can only travel very short distances before it disappears. This is called the "exciton bottleneck". To overcome this, the donor and acceptor materials are mixed into a combined nano-scale network, ensuring that an interface is always within reach of a generated exciton. The charge carrier mobility in these organic films is based on a "hopping" mechanism between molecules. The ability to fine-tune the chemical structure of the molecules allows for customised properties, such as transparency or specific colours, further increasing the technical potential of OPV.[14]

Because exactly these features were required, this specific type of PV was chosen for our project. It was necessary to achieve high light transparency while keeping the overall PV system lightweight.

3.3 Peak Solar Production

There are two types which are important for the peak solar production. On one hand, the peak solar power and on the other, the peak solar hours.

Regarding photovoltaic modules, peak solar power is defined as the power generated under ideal conditions. However, in the real world, environmental factors must be considered as they noticeably affect the performance. In the market, peak power is primarily used for the classification and comparability of panels. For example, a panel with 1 kWp (Kilowatt-peak) will produce up to 1 kW of electricity under ideal conditions.[29]

According to the source, the Standard Test Conditions (STC) for measurement are [29]:

- the ambient temperature at 25°C
- the irradiance at 1000 W/m²
- an Air Mass (AM) of 1.5, which represents the average solar position in mid-latitude regions

Another testing environment is the Nominal Operating Conditions (NOC), which simulates outdoor real-world conditions. The following values are assumed here[29]:

- the ambient temperature of 20°C
- the solar irradiance of 800 W/m²
- wind speed of 1.0 m/s at sea level

Nevertheless, it remains difficult to predict the exact electricity output of the modules, as it varies on changing environmental influences.[29]

There is also the concept of Peak Sun Hours (PSH), which indicates the duration of solar irradiance at a peak level of 1 kW/m². This happens when the modules reach their maximum potential. In the morning and evening, the solar angle is very small, and the light is less intense, resulting in lower power generation. As a result, the peak power is achieved during the middle of the day. The following formula can be used to calculate daily energy production[28]:

Daily Energy Production (kWh) = Solar Peak Power (kW) · Peak Sun Hours (PSH)

However, this formula does not account for the environmental factors that can affect the result. Such deviations can be caused by shading from trees, orientation, tilt angle of the panels, or weather conditions (seasonal variations). These examples apply to both peak sun hours and peak solar power.[28]

3.4 PV System Orientation

In this chapter, the different orientations of PV systems are evaluated based on their total energy yield, their production timeline throughout the day, and their overall benefit to the power grid. This study specifically focuses on two orientations: south and southeast. To illustrate the performance differences, both configurations are modelled and compared using a simulation.

3.4.1 South-facing

First, the traditional south-facing orientation will be analysed. Regarding the yearly energy yield, south-facing systems are the most productive. The production peak takes place between 11:00 AM and 2:00 PM, precisely during the midday peak. This is due to the sun reaching its highest point in the sky (solar noon).[1]

When comparing this production profile to typical household electricity consumption, a significant mismatch is observed. While the system generates maximum power at midday, the end-user demand usually peaks in the late afternoon and evening. Another noteworthy aspect is that the highest efficiency (Performance Ratio) is achieved in January, as PV modules operate more efficiently at lower temperatures. Conversely, the lowest efficiency is recorded in July and August due to the negative impact of high heat on the modules.[1]

3.4.2 Southeast-facing

In contrast, the southeast orientation produces slightly less total energy than the south-facing setup. However, the period of maximum production occurs significantly earlier, beginning at 09:00 AM and ending around 12:00 PM. In the afternoon, the power output drops off more rapidly.[1]

The efficiency (Performance Ratio) follows the same seasonal pattern as the south orientation, peaking in January and reaching its lowest levels in July and August. In terms of the grid infrastructure, this setup provides a clear advantage by reducing the midday peak, as the energy feed-in is shifted to the earlier morning hours.[1]

3.4.3 East-facing

An east-facing orientation generates a relatively high energy output in the morning, which can cover the early peak demand. However, the production already begins to

decrease by the early afternoon. Therefore, the total daily yield is lower compared to a south-facing orientation. Nevertheless, this setup is a practical solution for large-scale areas. Furthermore, the thermal stress on the modules is more uniform, which implies a longer lifespan. It should be noted, however, that fog during the spring and autumn months can negatively impact the performance.[6]

3.4.4 West-facing

In this orientation, the peak output takes place in the afternoon, whereas a south-facing orientation reaches its maximum at midday. Since energy consumption typically increases in the early evening, a west-facing orientation can be beneficial because of the matching load profile. Furthermore, this setup can improve the grid stability by reducing peak loads on the network when PV electricity is consumed directly by the user. However, this orientation generates less total energy because the angle of incidence is less favourable. Additionally, the specific location of the building is crucial, as some regions are at risk of fog or storms, which can impact performance.[5]

Chapter 4

Plant Selection and Growth Requirements

4.1 Requirements for Cultivation

For this project, the greenhouse is built directly on the ground without a prepared foundation. To create ideal soil conditions, approximately 50 cm of the existing ground should be replaced. The optimal soil structure should be as follows, moving from the top layer down:[12]:

Layer and Thickness	Material and Function
10 cm Potting soil	This soil should have a fine, crumbly texture and be rich in nutrients. It provides the optimal environment for planting.[12]
15 cm Garden soil	This layer ensures a loose structure that supports the transport of nutrients from the layers below and above.[12]
10 cm Foliage, shredded plant remains and coarse compost	This layer provides a high concentration of nutrients and essential soil microorganisms. It is strategically positioned at this depth to encourage plants to develop deeper and stronger root systems.[12]
Thin Layer Grass clippings	This is used as a separation between the upper and lower layers of the soil structure.[12]
15 cm Coarse cuttings (twigs and hedge trimmings)	By promoting air circulation and preventing water-logging, this airy layer helps to prevent root rot by optimising drainage.[12]

Table 4.1: Overview of the Layered Soil Structure

Soil maintenance is important. Therefore, it should be taken care of with organic fertilizer and the complete soil should be refreshed every three years.[11] The location should be chosen to ensure maximum sun exposure throughout all seasons while remaining sheltered from the wind[18], as plants require sunlight for photosynthesis. To maximise space, using multiple levels like shelves or hanging planters is recommended.[12] The soil texture should be loose and contain some clay particles to improve the flow of water and oxygen to the roots.[18] Moreover, proper ventilation is also essential to provide plants with fresh air and to regulate humidity and climate.[16] Finally, all these components work together to create the perfect environment for plant development.

4.1.1 Sample Year-Round Crop Plan

There are various types of vegetables, including climbing, fast-growing, and low-growing plants. Most of these are single-season crops and are mostly tolerant to direct sunlight.[12]

The following two images illustrate how seeding and planting can be structured during the spring and summer months.[16]

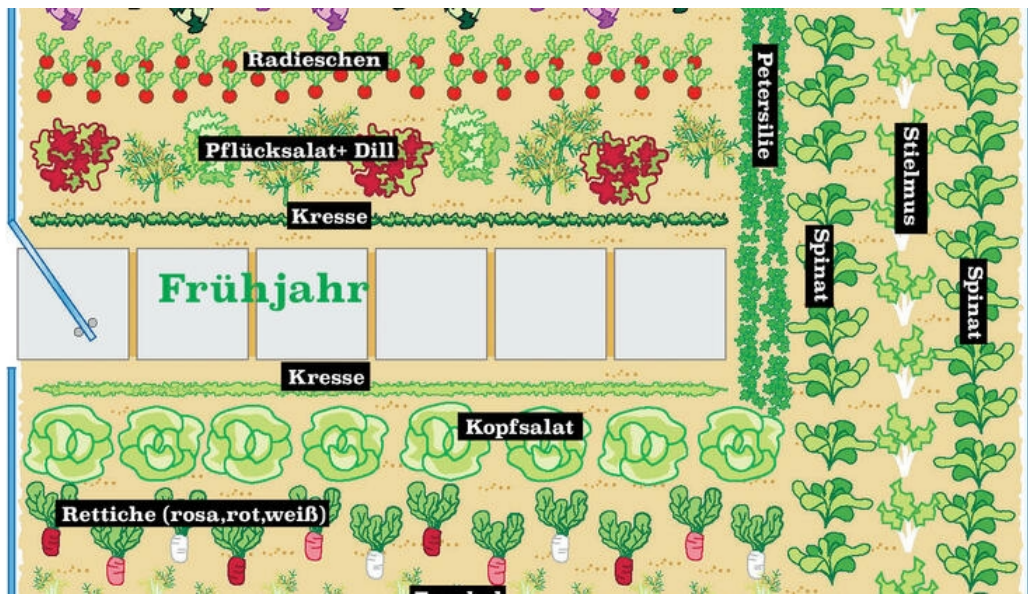


Figure 4.1: Spring Cultivation Plan[16]

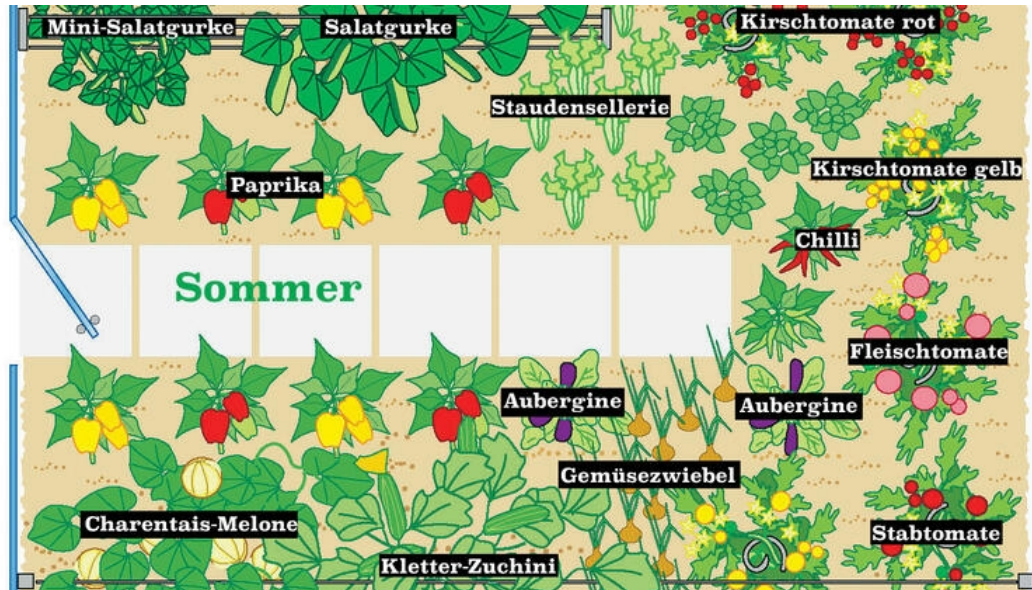


Figure 4.2: Summer Cultivation Plan[16]

In winter, leeks, various types of cabbage, and lamb's lettuce can be harvested. Spinach, leaf lettuce, and arugula follow in March. During spring, early carrots, radishes, and kohlrabi are available. Starting in May and continuing through the summer, the harvest season begins for many other vegetables, such as cucumbers and tomatoes. This period can extend into autumn. However, there are also specific autumn vegetables like endive, parsnips, pumpkin, and Chinese cabbage.[12]

Garden cress is an excellent choice for filling gaps in the beds. It grows the whole year.[11] For hanging baskets, strawberries are a great option, especially varieties that bear fruit multiple times throughout the year. This provides a year-round supply of fresh vegetables. Although it is necessary to follow seasonal patterns even in a warm greenhouse, the standard harvest periods can be extended.

Chapter 5

Implementation

5.1 A Python-based Simulation: Stochastic Modelling of Photovoltaic Output

The thesis uses a Python-based simulation to calculate and visualise the potential electricity production of a photovoltaic system. Moreover, the weather conditions play an important role in this model. Therefore, average values with stochastic variations are implemented. Furthermore, various approaches will be analysed. The first focus is on the orientation: three different orientations will be simulated and their electricity yields compared in graphs. Additionally, the impact of specific weather conditions will be discussed and contrasted with optimal weather scenarios. Finally, all components will be evaluated in combination to determine and explain the optimal configuration for the system.

5.2 Implementation

The implementation of this work aims to represent a full production year of the photovoltaic system, sourcing weather data from 2025. To optimise data reliability, I extracted meteorological information from three different weather websites for integration in my simulation, specifically focusing on:

- Maximum and minimum temperatures[21][8]
- Cloud cover[8]
- Daylight hours[27]

All weather data refer to the location of Klagenfurt.

The simulation is programmed and modulated in Python. The core code logic was adapted from the GreenCodesPython repository[22], which serves as a modern Python implementation of established photovoltaic algorithms originally written

in MATLAB from the book “Modeling of Photovoltaic Systems Using MATLAB: Simplified Green Codes”.[13]. Also, the Heliatek HeliaSol Datasheet was integrated into the code to ensure real-world accuracy.[10]

5.2.1 Setup and Initialization

The parameter tilt represents the angles at which the PV modules are installed. Since they are applied on different sides of the greenhouse, the angles are dissimilar. The first angle depends on the roof tilt, which is 16.26° . There are 5 PV panels installed on the roof at the first angle. The other two modules were attached to the vertical glass wall. These have an angle of 90° .

South is defined as 180 degrees, West as 270 degrees, and Southeast as 135 degrees. Furthermore, the manufacturer specifies the nominal power (peak power) between 0.055 kW and 0.060 kW per module. For my calculations, I assume an average value of 0.055 kW. The greenhouse is assumed to be located in Klagenfurt, so the weather data for this specific location is used. The weather data for Klagenfurt was obtained from three different sources. The temperature values of a daily minimum average and a daily maximum average were documented in an Excel table. Although the average daily minimum temperature were not involved in the code, because at night there can be no solar production. In addition, the duration of the sunshine was documented in the table and used in the code. The table containing the complete weather dataset can be found at my GitHub repo.[26]

5.2.2 Simulation Process

The simulation iterates through each day using an average. A comparative analysis displayed that increasing the temporal resolution to a daily basis showed no significant deviations from the reference-day approach.

Real-world weather is modelled using the sunshine ratio, a comparison between actual sunshine duration and the maximum potential daylight hours. The model assumes 100% performance during direct sunlight. Cloud cover is modelled by weighting the output at 10% to reflect the transition from direct to diffuse light.

Next, an hourly calculation is performed for the reference day. The simulation synchronises solar positioning with air temperature profiles, accounting for the natural phase shift known as thermal lag. The model derives the energy output from the density of irradiance, specifically on the surfaces. Finally, the simulation contrasts the real-world weather with the idealised clear-sky conditions.

5.3 Simulations

Based on the initial sketch, the photovoltaic system consists of a specific arrangement divided between two different installation surfaces. Five surfaces on the rooftop and

two surfaces on the wall (facade). A key characteristic is that the modules mounted on the wall are always positioned in the opposite direction of the primary rooftop orientation. For example, in a south-facing system, the wall-mounted modules are consequently oriented toward the north. The system tracks the energy yield of the roof and wall modules independently and combines them into an over one-year result. The simulation results are visualised in the following graphs.

5.3.1 Simulation 1: South

Based on my simulation parameters, the first graph presents the annual energy production of the rooftop system, and the second graph shows the production from the wall.

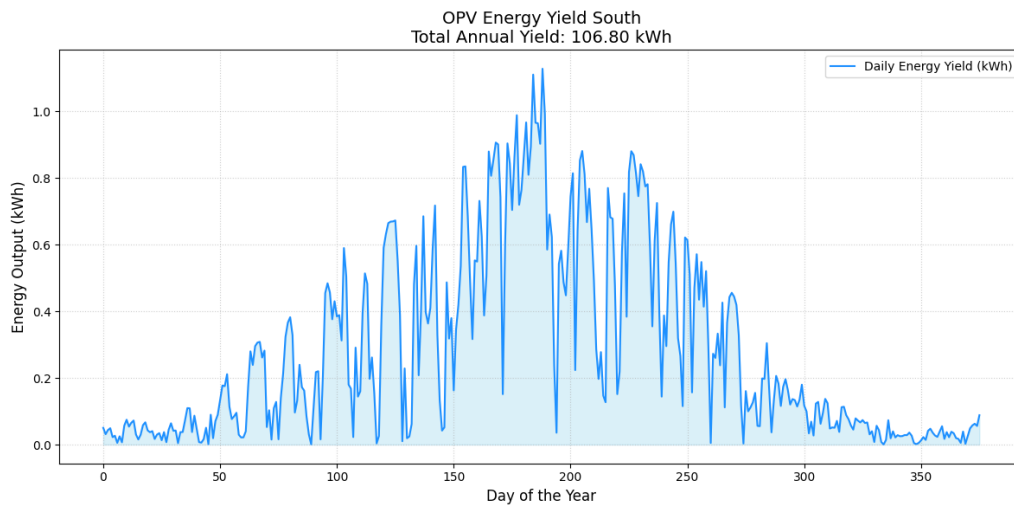


Figure 5.1: Annual yield South Roof

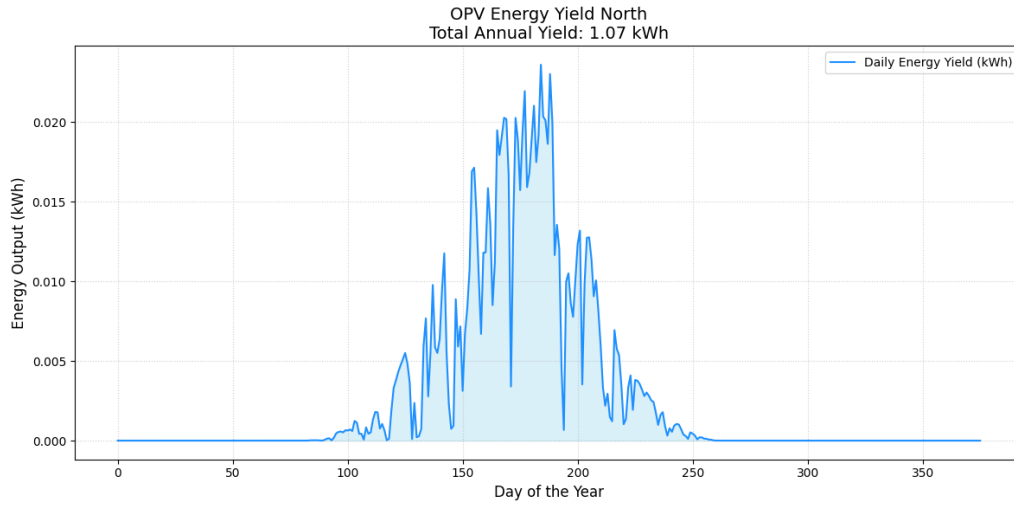


Figure 5.2: Annual yield North Wall

The last graph represents the combined annual energy production, integrating both the five rooftop modules and the two wall-mounted modules into a single plot.

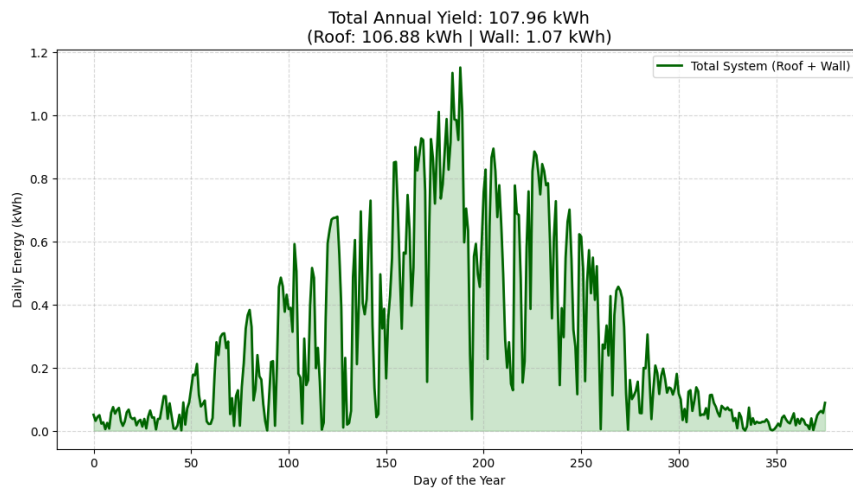


Figure 5.3: Combined Annual yield South and North

5.3.2 Simulation 2: Southeast

The same procedure goes for these parameters. The first graph presents the annual energy production of the rooftop system, and the second graph shows the production from the wall.

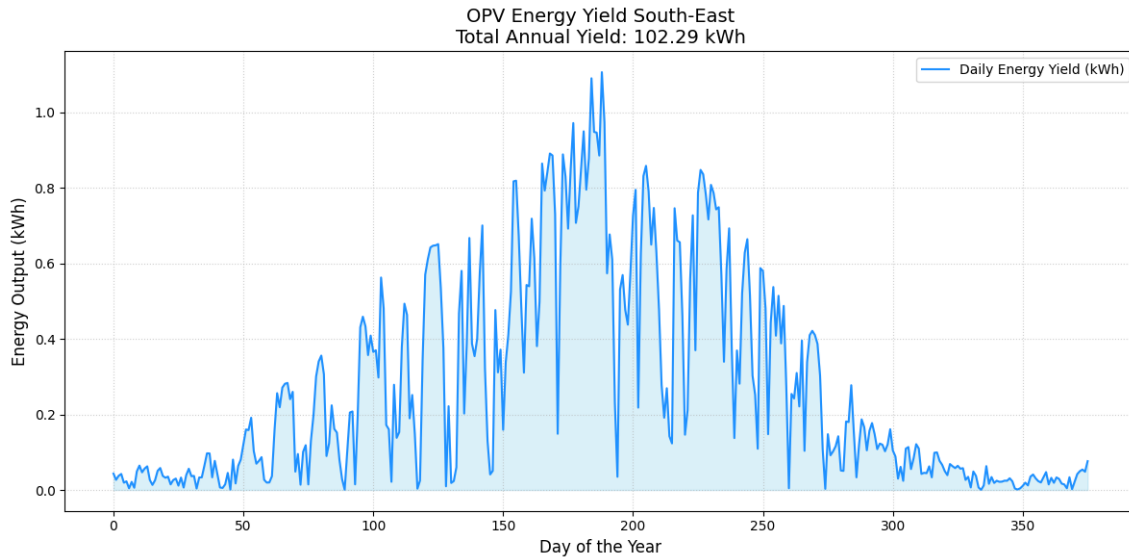


Figure 5.4: Annual yield Southeast Roof

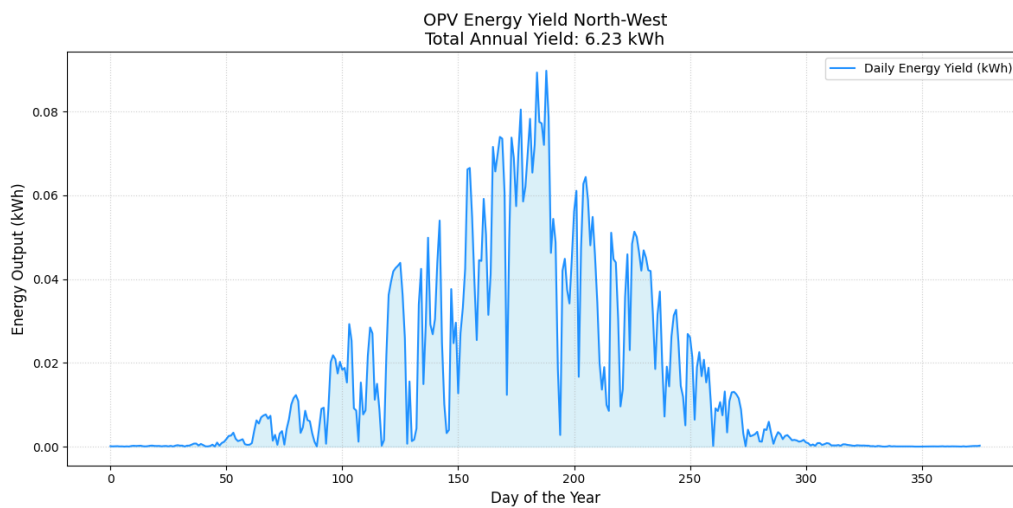


Figure 5.5: Annual yield Northwest Wall

The next graph illustrates the combined annual energy production, integrating both the five rooftop modules and the two wall modules into a single plot.

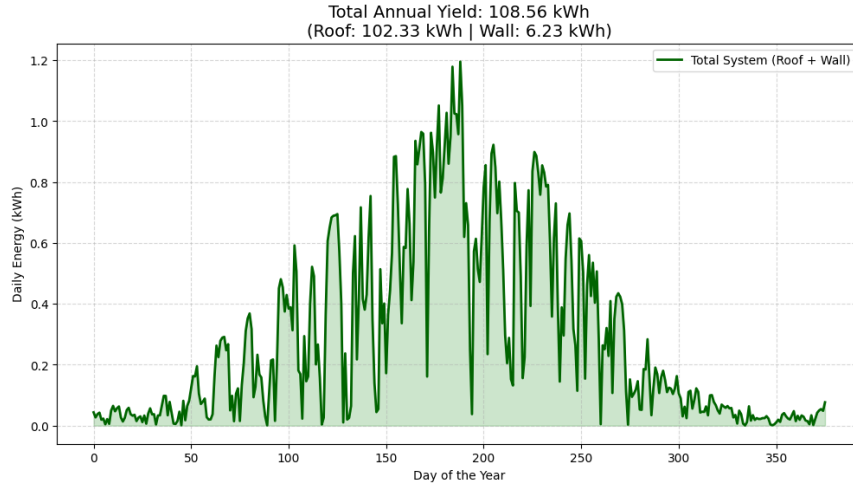


Figure 5.6: Combined Annual yield Southeast and Northwest

5.3.3 Simulation 3: East and West

The annual totals for East and West are identical because the sun's path is perfectly symmetrical. Since my simulation uses average values for weather and temperature, the morning conditions for an East panel mathematically mirror the afternoon conditions for a West panel. To see a difference in these results, I would need more precise, real-world measurement data. This would show characteristic patterns, like morning fog or hotter afternoons, which averages simply smooth out. Because of the aspects mentioned above, the combined annual production of East and West orientations is identical. Consequently, only a single representative graph is shown here.

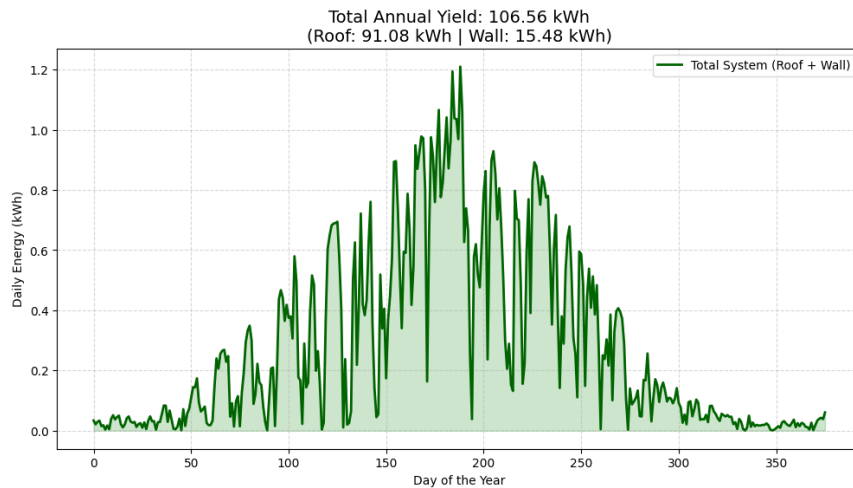


Figure 5.7: Combined Annual yield East and West

West

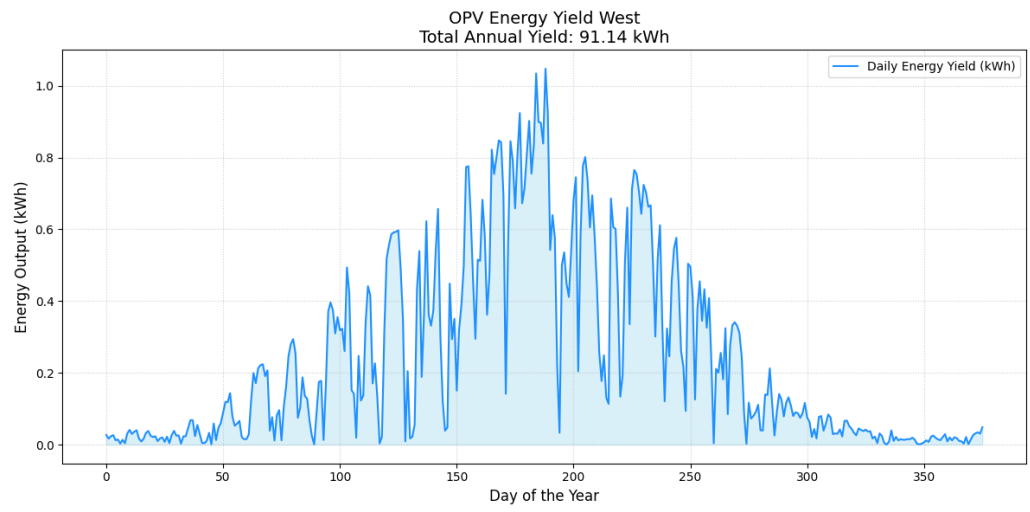


Figure 5.8: Annual yield West Roof

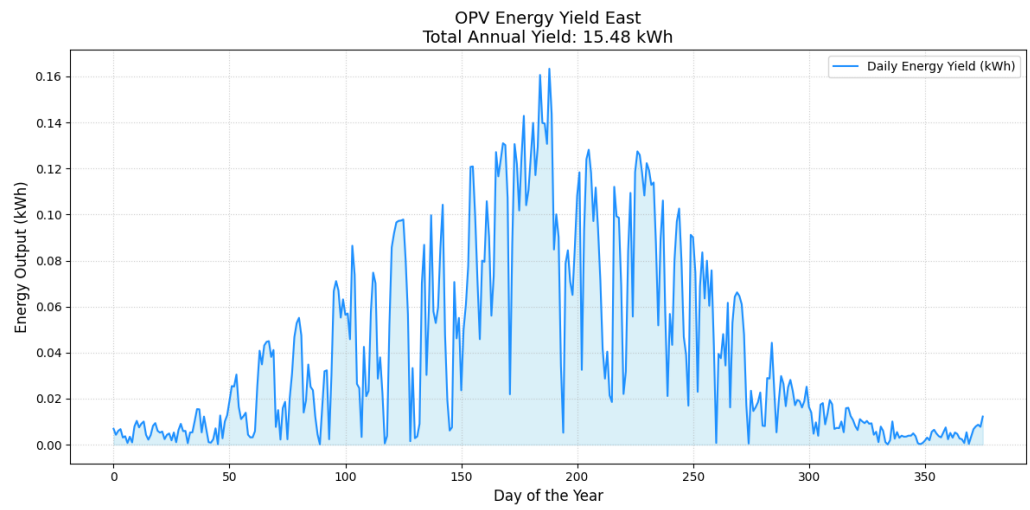


Figure 5.9: Annual yield East Wall

East

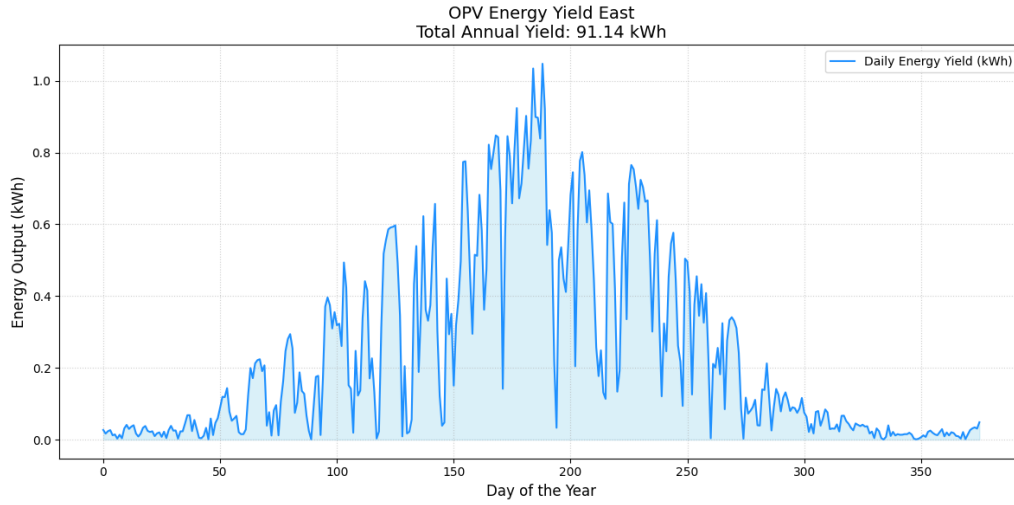


Figure 5.10: Annual yield East Roof

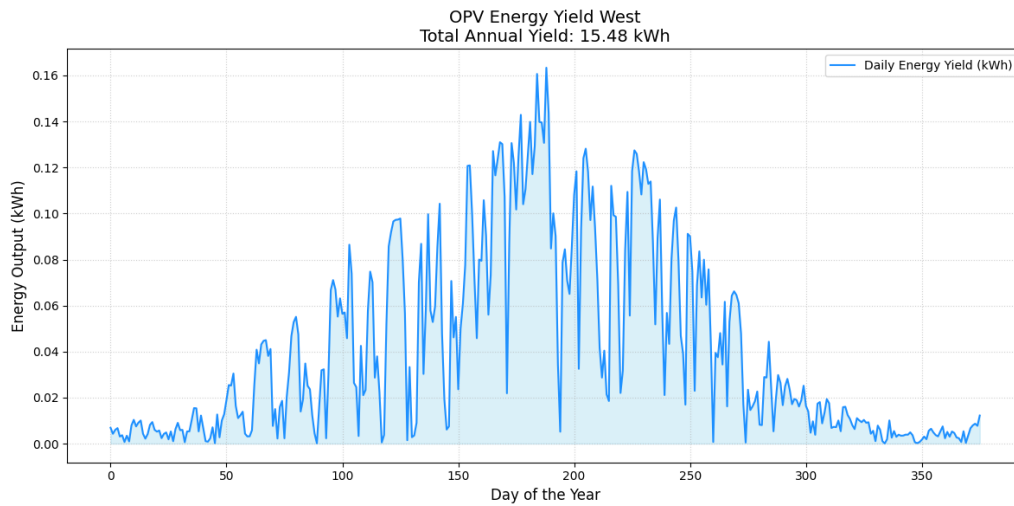


Figure 5.11: Annual yield West Wall

5.4 Results and Discussion

Results

- Annual yield with weather impact south: 107.96 kWh
- Annual yield with weather impact southeast: 108.56 kWh
- Annual yield with weather impact east and west: 106.56 kWh

The total annual result is a combination of the energy production of the OPV modules on the roof and those on the wall. It is important to mention that both systems are tracked separately with MPP-Tracker. Consequently, the performance of the wall modules directly affects the final total output. For instance, the North-facing wall generates only 1.07 kWh annually. Combined with the South-facing roof installation, they achieve a yearly yield of 107.96 kWh. This significantly impacts the overall result. While the Northwest orientation performs more efficiently, because of this the Southeast orientation results in the highest total annual output for the entire simulation.

As illustrated in the plot, the southeast-facing module outperforms the other orientations during spring and fall. However, during the winter, the south orientation shows the highest production peaks. Based on these results, I can conclude that for my specific location, a southern or southeast orientation is the most favourable and lucrative choice. During late spring, summer and early fall, the system maintains an output of a minimum of 0.6 kWh. This amount of power is sufficient to operate small fans, smart garden controllers for watering systems, small electronic devices, or LED lighting. Even an E-bike could be loaded, but this depends on the individual E-bike battery.

Furthermore, by deactivating the cloudiness parameter, I can calculate the specific energy loss caused by cloud cover, as demonstrated below. The corresponding graphs for the South-facing example are provided here.

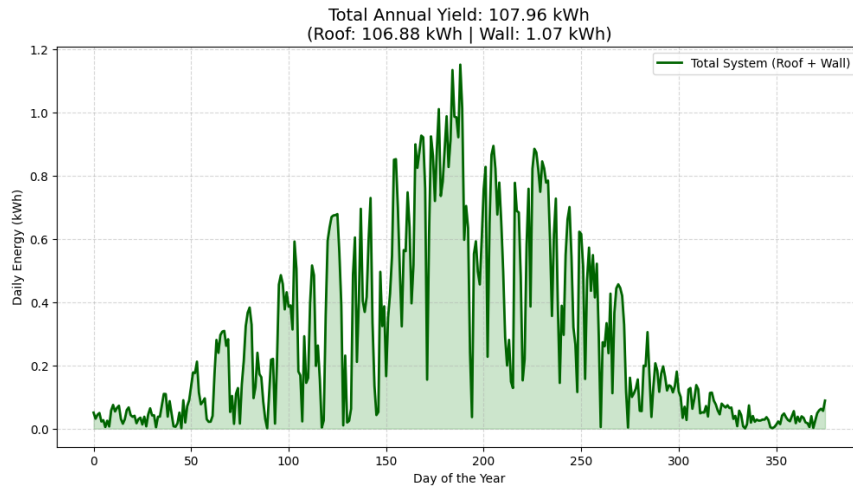


Figure 5.12: Combined Annual yield South and North with clouds

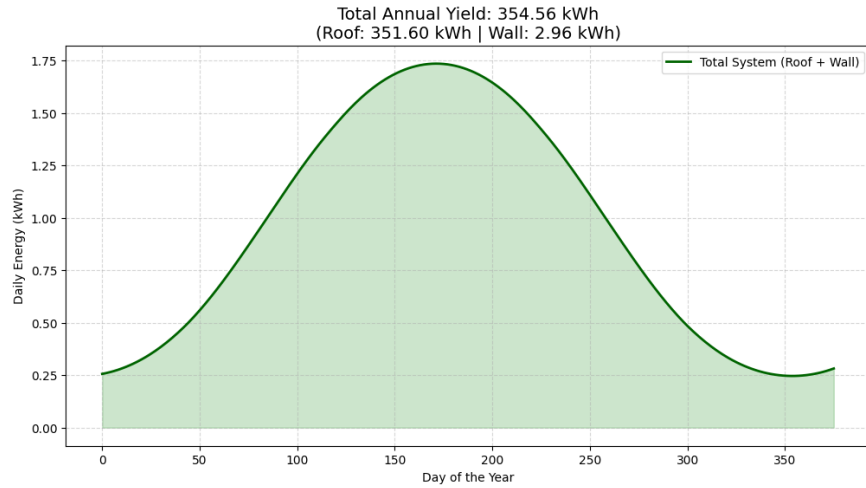


Figure 5.13: Combined Annual yield South and North ideal condition

Furthermore, the impact of cloud cover on electricity production is clearly visible. By implementing real-world weather data, I calculated an efficiency loss of approximately 69.55% compared to ideal conditions. For example: 0.86 kWh are 100% the ideal. The relative yield is calculated as $\frac{107.96}{354.56} \times 100 = 30.44\%$. Now I can calculate the efficiency loss caused by external factors. The loss is a round 69.55%.

Chapter 6

Conclusion

6.1 Summary

Based on costs and with a view to sustainability, it can be concluded that this project is designed for long durability, which justifies the initial investment for me. Furthermore, the majority of the components used are eco-friendly and recyclable. The only exception is the watering system, which is currently not sustainable. Unfortunately, no small-scale watering system was found that operates without batteries. Otherwise, it could have been powered directly by the self-generated electricity.

My theoretical research regarding the different orientations is similar to the simulation results. The southern and southeast orientation proved to be the most efficient for energy production over the year. Therefore, I chose this orientation for the final evaluation of a potential greenhouse deployed in Klagenfurt. Regarding the cultivation process, there are different approaches. My theoretical planting plan would need to be verified through a real-world simulation, which would likely allow for further optimisation of the system.

The watering controller is programmable. Based on my own experience, I would initially set the system to run for one hour in the morning and one hour in the evening. However, a real-world simulation would be beneficial to determine the most effective timing. These aspects could be further tested and proven in future studies.

In conclusion, this project represents a user-friendly introduction to a solar-supported greenhouse. It is recommended for anyone who is looking to produce their own electricity for garden use.

6.2 Further Work

There are different ways for further research on this study. I have outlined two specific areas of interest that could expand the reach of this project. Both are based on a real-world simulation of the greenhouse. The first approach focuses on a

more detailed analysis of the electrical data collected, while the second explores the biological aspects of the system.

6.2.1 Real-life Experiment

To deepen my research, the next step would be building an actual greenhouse to compare my current results and real-world measurements. While my values are based on ideal scenarios, including a slight tolerance, a physical model would provide empirical results. In addition, factors such as long-term durability could be tested. This approach focuses on the greenhouse itself and its energy production efficiency.

6.2.2 The Biological Aspects

A second approach could focus on the biological side of the project. The goal would be to evaluate exactly which plant varieties are best suited for year-round harvesting. Furthermore, it is important to explore how the greenhouse design can be optimised for the plants growing inside. Another key element could be determining what additional tools are needed to boost plant growth or integrate smart features to make the entire system more autonomous. This could also be evaluated with a real-life greenhouse.

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